

ShopSphere E-Commerce

Data Analysis & Power BI Dashboard

Complete Project Report — From Data Inspection to Business Insights

Dataset	ShopSphere E-Commerce
Rows	200,000
Columns	24
Unique Orders	200,000
Unique Customers	49,054

Project objective: Analyze ShopSphere transaction data to understand sales performance, customer behavior, product/category performance and profitability, then present the findings through an interactive Power BI dashboard.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn and Microsoft Power BI.

1. Business Problem & Project Objectives

The project was designed to turn a large e-commerce transaction dataset into clear business information that management can use to understand performance and identify opportunities.

- Measure overall sales, orders, customers, average order value and profitability.
- Understand monthly sales trends and compare 2024 with 2025.
- Identify which product categories contribute most to sales.
- Check data quality before building the dashboard.
- Create a clean, management-friendly Power BI dashboard.
- Convert analysis into actionable business insights and recommendations.

2. Dataset Overview

The source dataset contains **200,000 rows and 24 columns**. Each row represents an e-commerce transaction/order record. There are **200,000 unique Order_IDs** and **500 unique Product_IDs**.

Group	Columns
Order & customer	Order_ID, Order_Date, Customer_ID, Customer_Name, Age, Gender, City
Product	Product_ID, Product_Name, Brand, Category
Transaction	Quantity, Unit_Price, Discount, Revenue, Discount_Amount, Final_Sales, Cost, Profit, Profit_Margin
Operations	Payment_Method, Order_Status, Shipping_Days, Rating

All 24 column names: Order_ID, Order_Date, Customer_ID, Customer_Name, Age, Gender, City, Product_ID, Product_Name, Brand, Category, Quantity, Unit_Price, Discount, Revenue, Discount_Amount, Final_Sales, Cost, Profit, Profit_Margin, Payment_Method, Order_Status, Shipping_Days, Rating.

3. Data Inspection & Data Quality Checks

The analysis started with Pandas inspection: previewing records, checking shape, column names, data types, missing values, duplicate rows and statistical summaries.

Check	Result	Interpretation
Rows	200,000	Full transaction count
Columns	24	24 fields available for analysis
Missing values	0	No missing values
Duplicate rows	0	No fully duplicated transaction rows
Unique Order_ID	200,000	Order IDs are unique
Unique Customer_ID	49,054	Customer base identified
Unique Product_ID	500	500 unique products

Missing Values by Column

Column	Missing	Column	Missing
Order_ID	0	Order_Date	0
Customer_ID	0	Customer_Name	0
Age	0	Gender	0
City	0	Product_ID	0
Product_Name	0	Brand	0
Category	0	Quantity	0
Unit_Price	0	Discount	0
Revenue	0	Discount_Amount	0
Final_Sales	0	Cost	0
Profit	0	Profit_Margin	0
Payment_Method	0	Order_Status	0
Shipping_Days	0	Rating	0

4. Customer Repetition Analysis

The dataset contains 49,054 unique customers across 200,000 transaction rows. Repeated Customer_ID values are expected and are useful for understanding repeat purchasing behavior.

Customer metric	Count
Unique customers	49,054
Customers with >1 order	45,377
Customers with exactly 1 order	3,677
Additional repeat-order rows	150,946
Maximum orders by one customer	14

Interpretation: 45,377 customers appear more than once, indicating a substantial repeat-customer base. This supports the use of customer-level KPIs such as distinct customers and AOV rather than simply counting customer rows.

5. Numerical Validation & Consistency Checks

Validation	Result
Expected Revenue = Quantity × Unit_Price	0 mismatches
Expected Final_Sales = Revenue – Discount_Amount	60 mismatches before tolerance check
Expected Profit = Final_Sales – Cost	0 mismatches
Negative numeric values	0 negative values
Rows with zero discount	217

A small number of Final_Sales records differed from the direct calculation at a strict rounding comparison; a tolerance check was then applied. Profit was consistent within the defined 0.01 tolerance.

Order Status Distribution

Order Status	Orders	Share
Delivered	176,107	88.05%
Cancelled	13,958	6.98%
Returned	9,935	4.97%

6. Power BI Data Modeling & Measures

After Python-based inspection and validation, the dataset was brought into Power BI for interactive analysis and dashboard creation.

Modeling component	What was completed
DateTable	Created a dedicated date table for time-based analysis.
Date fields	Date, Month Name, Month Number, Year and Year Month.
Sorting	Month Number supports chronological sorting of Month Name.
Trend field	Year Month supports monthly analysis across 2024 and 2025.

Measures / KPIs Used

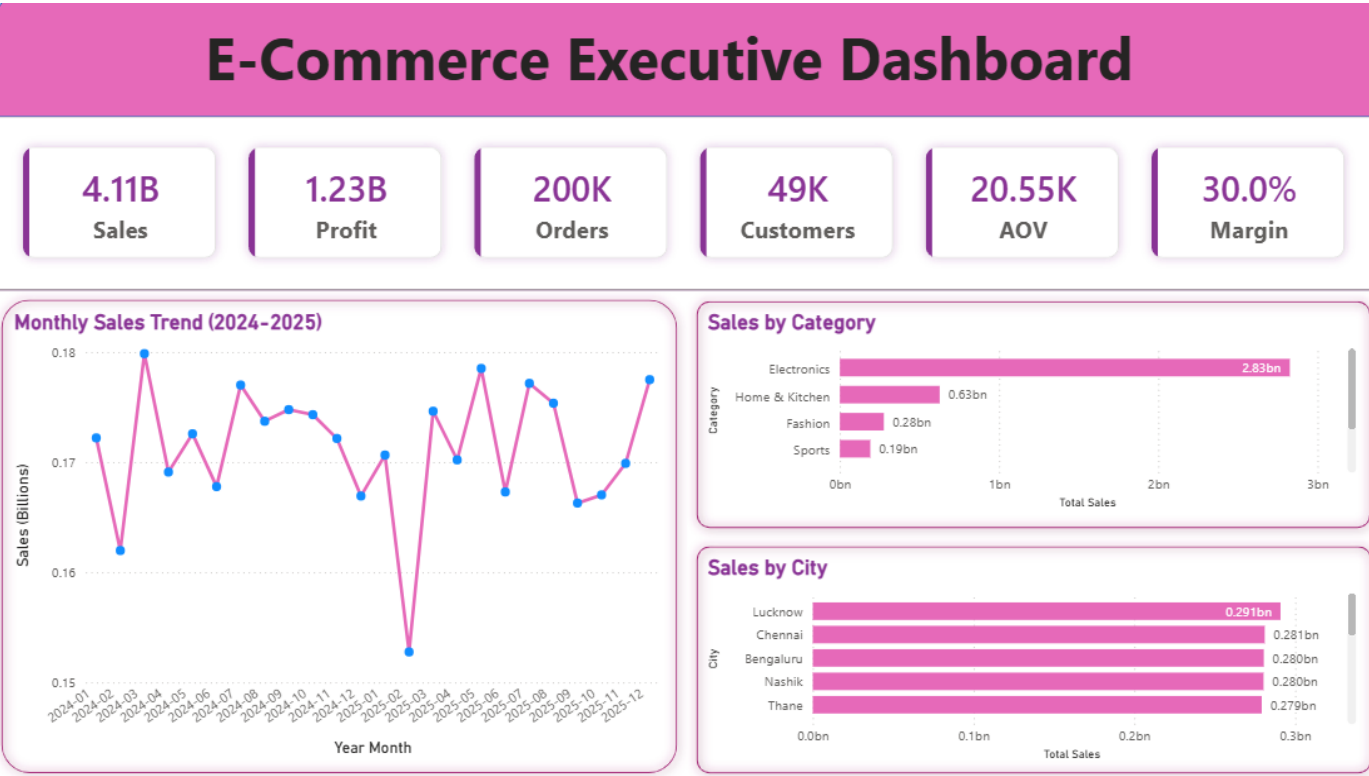
- Total Sales — total Final_Sales.
- Total Orders — order count based on the transaction dataset.
- AOV — Total Sales divided by Total Orders.
- Total Customers — distinct Customer_ID count.
- Profit and Margin — used in the executive view.

Dashboard Structure

The final Power BI work was organized into two pages: an **E-Commerce Executive Dashboard** for high-level business monitoring and a **Sales Performance Analysis** page for deeper sales comparison and category analysis.

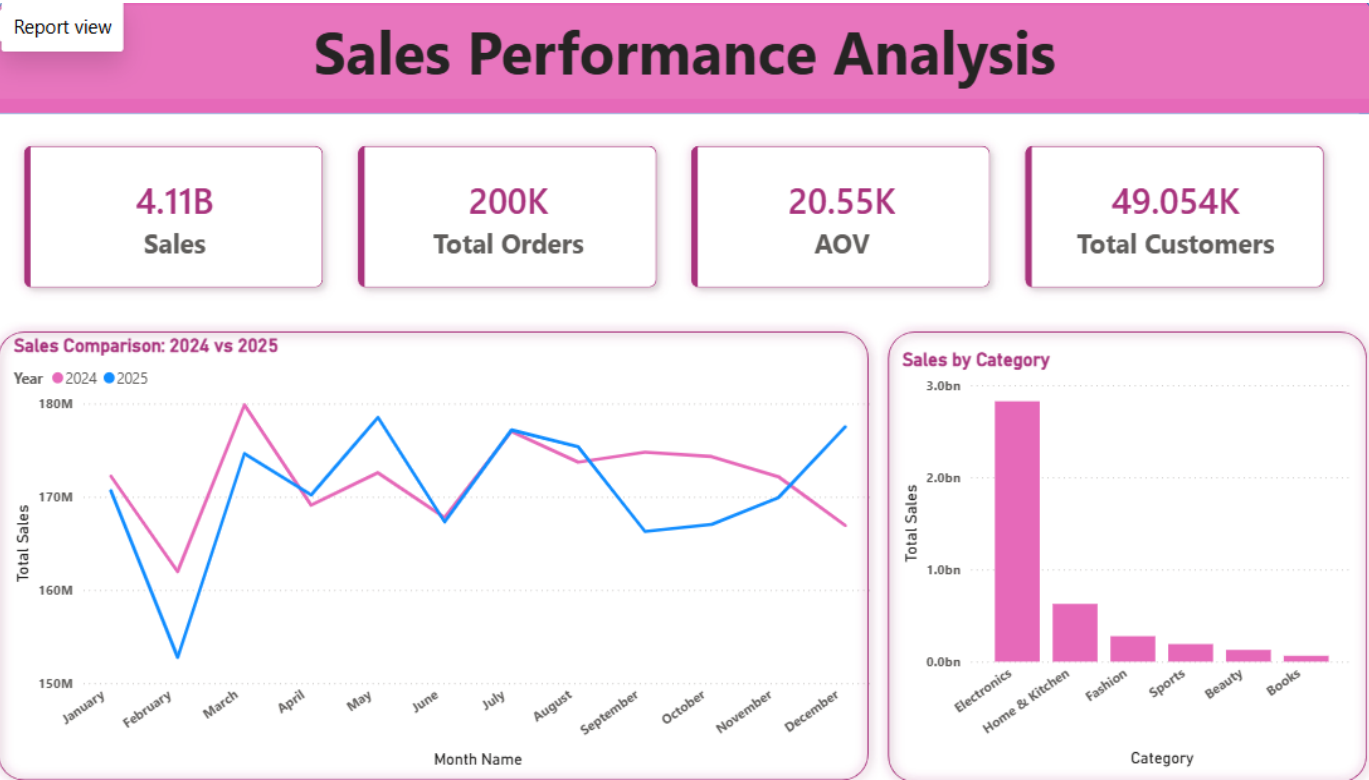
7. Power BI Dashboard — E-Commerce Executive Dashboard

The executive page combines KPI cards with monthly sales trend, category performance and city-level sales.



8. Power BI Dashboard — Sales Performance Analysis

The second page focuses on the 2024 vs 2025 monthly sales comparison and category contribution.



Visuals: KPI cards, Sales Comparison 2024 vs 2025 line chart and Sales by Category bar chart.

9. Business Insights

- The dataset contains 200,000 transactions and 49,054 distinct customers, showing a large transaction base.
- 45,377 customers have placed more than one order, indicating strong potential for retention and repeat-purchase analysis.
- Total Final_Sales in the source data is approximately 4.11B, with approximately 1.23B in profit.
- Overall AOV is approximately 20.55K, while the overall profit margin is approximately 30.0%.
- Electronics is the strongest sales category in the Power BI category visualization, followed by Home & Kitchen; the remaining categories contribute smaller shares.
- Monthly sales fluctuate across the two-year period, so management should monitor both high-performing periods and noticeable declines rather than relying only on annual totals.

10. Business Recommendations

- Protect Electronics: maintain availability of high-performing products and use targeted promotions to defend its leading contribution.
- Grow Home & Kitchen: investigate products and customer segments that can move this category closer to the top category.
- Improve repeat purchasing: segment the repeat-customer base and build personalized offers, bundles and loyalty campaigns.
- Use AOV strategically: introduce cross-sells and product bundles to increase order value without relying only on discounts.
- Monitor monthly performance: investigate sharp declines quickly and identify what caused strong months so successful patterns can be repeated.
- Balance growth with profitability: evaluate promotions using both sales and profit impact, not sales volume alone.
- Keep data validation in the refresh process: repeat duplicate, missing-value and financial consistency checks whenever new transaction data is loaded.

11. End-to-End Project Workflow

Stage	What was completed
1. Business understanding	Defined the e-commerce business problem and analysis objectives.
2. Data loading	Loaded the 200K ShopSphere transaction dataset with Pandas.
3. Initial inspection	Checked rows, columns, column names, data types and sample records.
4. Data quality	Checked missing values, duplicate rows, duplicate Order_IDs and numeric validity.
5. Customer analysis	Calculated distinct customers and identified repeated Customer_IDs.
6. Financial validation	Validated Revenue, Final_Sales and Profit calculations.
7. Power BI modeling	Created DateTable and prepared fields/measures for reporting.
8. Dashboard creation	Built Executive Dashboard and Sales Performance Analysis pages.
9. Visualization formatting	Applied KPI cards, line chart, bar charts and consistent pink styling.
10. Business interpretation	Converted dashboard patterns into insights and recommendations.

Final outcome: The project demonstrates an end-to-end data analytics workflow — from raw e-commerce transaction data and Python validation through Power BI modeling, visualization, business insights and recommendations.

Report note: The two dashboard visuals in this report use the Power BI images provided for the final project, replacing the earlier dashboard screenshot placeholders.